Enhanced Photocatalytic Activity of TiO_2 Anatase

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Abstract
We investigate the band gap engineering of TiO_2 anatase nanoparticles through nitrogen doping. DFT calcu

Introduction
Titanium dioxide remains the most widely studied photocatalyst due to its stability, low cost, and non-toxicity

Methods
DFT calculations were performed using VASP with PAW pseudopotentials. A 2x2x1 supercell of anatase Ti

Results
The calculated density of states shows a clear band gap reduction upon nitrogen incorporation.

Conclusion
Nitrogen-doped TiO_2 anatase shows significantly enhanced visible-light photocatalytic activity, validated by